Solving the static Einstein equations with physics-informed neural networks

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A classical existence theory guarantees that the stationary vacuum Einstein equations, posed in the region outside a ball, have a unique solution for boundary data sufficiently close to flat space, depending continuously on the data. We show that this boundary-value problem can be solved numerically with physics-informed neural networks (PINNs). We work with the static equations in harmonic gauge, written for a Riemannian 3-metric h and the squared norm λ of the static Killing vector. Data are a function λ and an induced metric on an inner sphere, together with asymptotic flatness imposed through Robin conditions on an outer sphere. Metric-only networks, in which the connection is the Christoffel symbol of the network metric, are three orders of magnitude more accurate than first-order networks with an independent connection. The solver reproduces Schwarzschild in harmonic coordinates to 10^{-4} and two-rod Weyl solutions to 10^{-7}. For a rotated Weyl configuration with no symmetry left, the error is 10^{-3}–10^{-4}. For quadrupolar data, which have no closed-form solution, we compare with an independent spectral Gauss–Newton solution. There we identify a residual harmonic gauge mode: a translation of the asymptotic chart that the boundary data do not fix. It dominates the error, at 0.062 times the centre-of-mass offset. We fix it with a centre-of-mass condition built from a conserved flux, the static analogue of a dipole moment. This reduces the error by four to five orders of magnitude, to five to six orders below the size of the quadrupole perturbation itself. We extract the Geroch–Hansen mass moments of the computed solutions from the symmetry axis, without constructing Weyl coordinates. The networks reproduce the exact moments of the Weyl solutions, and the reference moments of the quadrupolar solution ($M_0 = 0.583847$, $M_2 = 0.1747$, $M_4 = 0.022$) to 10^{-8}–10^{-4}. Even runs without the centre-of-mass condition agree to 10^{-5}: the translation mode moves the chart, not the geometry. We discuss the optimisers that make this possible: a self-scaled Broyden quasi-Newton method and a doubly-sketched Gauss–Newton method. We also discuss the limits we found for the general three-dimensional ansatz.

I. INTRODUCTION

For many years there has been a mathematical theory of the stationary Einstein equations in an asymptotic region of spacetime, namely the region outside a ball [1]. It states that boundary data close enough to those of flat spacetime, together with the asymptotic conditions, produce a unique solution that depends continuously on the data. Very few attempts have been made to explore this theory numerically [2]. Here we show that this is possible, and that it opens a new window on these spacetimes.

The method also gives a simple way to explore other systems of possible physical importance: systems in which one solves for a Riemannian metric together with other fields, through equations that contain its Ricci tensor. Examples are the construction of minimal-energy data sets [3] and the equations that give higher-order Newtonian-limit expansions [4]. [TODO: more examples]

We restrict this work to the static Einstein equations. The stationary case is no harder from the point of view of the approximation method, and is left for a continuation of this work.

Physics-informed neural networks [5] represent the unknown fields by a neural network and minimise the squared residual of the equations and boundary conditions over sampled points. They need no mesh, handle a non-compact domain through a change of radial variable, and treat a configuration with no symmetry at all exactly as they treat a symmetric one. Their weakness is accuracy. The loss is a mean square, the optimisation is non-convex, and a small loss does not by itself mean a small error. A large part of this paper is about how far that weakness can be pushed back for a geometric problem, and about the one place where it turned out to hide a structural ambiguity of the problem rather than a numerical one: the centre-of-mass gauge.

The paper is organised as follows. Section II states the equations, the gauge, the boundary and asymptotic conditions, and the residual gauge freedom. Section III describes the networks, the loss and the optimisers. Section IV validates the solver on exact solutions: Schwarzschild, and Weyl two-rod solutions on the axis and rotated. Section V treats quadrupolar data, the translation ambiguity and its fix, and the far field. Section VI concludes.

II. THE STATIC EINSTEIN EQUATIONS

A. Variables and field equations

We write a static vacuum spacetime as

$$g = -\lambda\, \mathrm{d}t^2 + \lambda^{-1} h, \tag{1}$$

where h is a Riemannian metric on the spatial slice and $\lambda = -\xi \cdot \xi$ is the squared norm of the static Killing vector $\xi = \partial_t$. If $g = -N^2 \mathrm{d}t^2 + \gamma$, then $\lambda = N^2$ and $h = N^2 \gamma$.

The vacuum equations become a system for (h, λ):

$$R_{ab}(h) = \frac{1}{2\lambda^2}\,\nabla_a\lambda\,\nabla_b\lambda, \tag{2}$$

$$h^{ab}\nabla_a\nabla_b\lambda = \frac{1}{\lambda}\,h^{ab}\nabla_a\lambda\nabla_b\lambda. \tag{3}$$

With $U = \frac{1}{2}\ln\lambda$ these read $\mathrm{Ric}(h) = 2\,\mathrm{d}U\otimes\mathrm{d}U$ and $\Delta_h U = 0$. The divergence of (2) reproduces (3) through the contracted Bianchi identity. Both equations are invariant under $\lambda \to c\lambda$ and $\lambda \to 1/\lambda$.

B. Gauge

We impose harmonic (de Donder) coordinates on h:

$$\Gamma^i \equiv h^{jk}\Gamma^i{}_{jk} = S^i, \tag{4}$$

with $S^i = 0$ in the harmonic case. Equivalently $\Delta_h x^i = 0$: the Cartesian-like coordinates $x^i = \rho\, n^i$ are h-harmonic. A polar chart can never satisfy this condition. For the Weyl tests of Sec. IV we use a source built from the candidate's own metric,

$$S^i = (h_{\rho\rho} - 1)\,h^{ij}\partial_j \ln\rho_n, \qquad \rho_n = \big|x - (x\cdot n)\,n\big|, \tag{5}$$

where n is the symmetry axis. This source is satisfied exactly by Weyl's canonical form $h = e^{2k}(\mathrm{d}\rho^2 + \mathrm{d}z^2) + \rho^2\mathrm{d}\phi^2$. We also implemented the DeTurck (harmonic-Einstein) reduction $R_{ij} - \nabla_{(i}\xi_{j)}$ with $\xi^k = \Gamma^k - S^k$, whose principal part is the elliptic operator $-\frac{1}{2}h^{cd}\partial_c\partial_d h_{ij}$ [6]. In our runs it gave no clear gain over imposing (4) as a separate residual (same loss trajectory, 1.21×10^{-10} against 1.16×10^{-10}).

C. The exact spherical family

In the harmonic chart, the spherically symmetric solutions are

$$h_{ij} = \Big(1 - \frac{R_0^2}{\rho^2}\Big)\delta_{ij} + \frac{R_0^2}{\rho^2}\,n_i n_j, \qquad \lambda = k\,\frac{\rho - R_0}{\rho + R_0}, \tag{6}$$

so that $h = \mathrm{d}\rho^2 + (\rho^2 - R_0^2)\,\mathrm{d}\Omega^2$. In the reading (1) with $k = 1$ this is Schwarzschild in harmonic coordinates: the mass is $M = R_0$, the areal radius is $r = \rho + R_0$, and $\lambda = 1 - 2M/r$. The harmonic condition still allows the radial maps $\rho \to F(\rho) = c_1\rho + c_2 F_2(\rho)$ with $F_2 = 2R_0 + \rho\ln\frac{\rho - R_0}{\rho + R_0}$. Both solve $(f^2F')' = 2F$ with $f^2 = \rho^2 - R_0^2$. The round inner metric and $h \to \delta$ fix both constants.

D. Boundary and asymptotic conditions

The domain is the shell $\rho_{\mathrm{in}} \le \rho \le \rho_{\mathrm{out}}$.

a. Inner sphere. We prescribe λ and the induced metric:

$$\lambda = \lambda_0 + S_1\, n\cdot e + S_2\, P_2(n\cdot e), \qquad PhP = \frac{r_{\mathrm{in}}^2}{\rho_{\mathrm{in}}^2}\,P, \tag{7}$$

where $P = I - n\otimes n$, r_{in} is the prescribed areal radius of the sphere in h, and e is a fixed unit vector (the data axis). The components h_{rr} and h_{rA} are *not* data. An earlier version also imposed $h_{rA} = 0$, which over-determines the problem. In harmonic gauge the $l = 2$ sector has five second-order radial equations, hence five decaying constants, and the datum (7) with the gauge already fixes five. A sixth condition leaves no decaying solution, and the spectral solver of Sec. III D indeed stalled at a residual of 1.8×10^{-5} with a non-decaying $l = 2$ metric part until it was removed. The constant λ_0 is chosen so that $\lambda \to 1$ on the spherical branch, $\lambda_0 = (\rho_g - R_0)/(\rho_g + R_0)$ with $\rho_g = \sqrt{r_{\mathrm{in}}^2 + R_0^2}$.

b. Outer sphere: Robin conditions. Asymptotic flatness is imposed at ρ_{out} through Robin conditions of order n,

$$B_n[f] \equiv \prod_{i=0}^{n-1}\big(\rho\,\partial_\rho + p_f + i\big)\,(f - f_\infty) = 0, \tag{8}$$

which annihilate the decaying powers $\rho^{-p_f},\dots,\rho^{-p_f-n+1}$ exactly. The exponents follow from the equations: $\delta h \sim \rho^{-2}$, $\Gamma \sim \rho^{-3}$ and $\lambda - 1 \sim \rho^{-1}$ (measured ratios on the exact solution: 2.01, 3.03 and 0.955). Writing the higher-order condition directly in ∂_ρ instead of $\theta = \rho\partial_\rho$ is wrong; the factors of (8) commute. For the scalar we impose the condition on the Geroch–Hansen potential [7, 8],

$$\mathcal{P} = \frac{1 - \lambda^2}{4\lambda} = -\tfrac{1}{2}\sinh 2U, \tag{9}$$

rather than on λ. The reason is that the monopole of $\mathcal{P}$, $M/\rho + M^3/(3\rho^3) + \dots$, has no ρ^{-2} term, whereas $\lambda = 1 - 2M/\rho + 2M^2/\rho^2 - \dots$ has one, and the order-one condition on λ is violated by the exact solution (Sec. V). The same condition is imposed on a ladder of spheres inside ρ_{out} (at 0.7, 0.8 and $0.9\,\rho_{\mathrm{out}}$). The condition is homogeneous and the exact solution satisfies it at every radius. Sampled at a single radius it was "gameable": $B_3[h - I]$ was 1.30, 0.52 and 0.059 at $\rho = 50$, 90 and 99 but 4.8×10^{-5} at $\rho = 100$, while the bulk kept a constant $l = 2$ metric part of about 6%.

c. Averaged pins. Spherical means of order-one Robin combinations are added at ρ_{out}: $\langle(\rho\partial_\rho + 3)(h_{rr} - 1)\rangle = 0$, $\langle(\rho\partial_\rho + 2)(g_2 - 1)\rangle = 0$ with $g_2 = \frac{1}{2}(\mathrm{tr}\,h - h_{rr})$, and $\langle(\rho\partial_\rho + 1)(\mathcal{P} - \mathcal{P}_\infty)\rangle = 0$. Without the metric pins h drifts towards a constant multiple of the identity. $\lambda_\infty = 1$ is fixed; if it is left free, the trivial branch $h = \delta$, $\lambda = \mathrm{const}$ is reached.

E. Uniqueness and the residual gauge freedom

The harmonic condition (4) fixes the chart only up to harmonic diffeomorphisms $x \to x + \xi$ with $\Delta_h\xi = 0$. One

of these is not fixed by our data. It is the solution with $\xi = 0$ on the inner sphere and $\xi \to a$ at infinity: it keeps the chart harmonic, leaves the inner data unchanged, and translates the asymptotic chart by a constant vector a. Its effect on the scalar, $\xi \cdot \nabla\lambda$, vanishes at $\rho_{\rm in}$, peaks near $\rho \approx 2$ and decays like ρ^{-2}. The outer conditions penalise it only weakly. For $|a| \approx 0.04$ the dipole of $\mathcal{P}$ at $\rho_{\rm out} = 100$ is $\approx 2.5 \times 10^{-6}$, which costs $\approx 10^{-9}$ in the loss. In the language of multipole moments, a mass dipole M_1 is pure gauge and a centre-of-mass condition sets it to zero [7–9].

To fix it we use a conserved flux. Since $U = \frac{1}{2}\ln\lambda$ and the coordinates x^i are both h-harmonic, Green's identity makes

$$D^i(S) = \oint_S \left(x^i\,\partial_n U - U\,\partial_n x^i\right) \mathrm{d}A_h \tag{10}$$

independent of the closed surface S enclosing the inner sphere. For a mass M displaced by a ($U \sim -M/|x-a|$), $D = 4\pi M a$. This is the pseudo-Newtonian centre of mass that Cederbaum introduced for static ("geometrostatic") isolated systems [10], written in our variables: $D^k = 4\pi m z^k$ in harmonic coordinates of the conformal metric, with the surface independence following from Green's formula exactly as here. It is quasi-local (it can be evaluated on any surface enclosing the sources), it obeys an addition theorem for several separated sources, it coincides with the ADM and constant-mean-curvature centres of mass in suitable asymptotic coordinates, and it converges to the Newtonian centre of mass in the Newtonian limit [10, 11]. The asymptotic notions it is compared with go back to Regge–Teitelboim and Beig–Ó Murchadha [12, 13], and to the foliation-based definitions of Huisken–Yau, Metzger, Huang, Corvino–Wu and Eichmair–Metzger [14–19]; see Ref. [20] for a review. Cederbaum's analysis builds on the asymptotics of static fields of Beig, Beig–Simon and Kennefick–Ó Murchadha [21–23]. We impose $d \equiv D/4\pi = 0$ on a set of spheres. This is a gauge choice: it selects the asymptotic chart centred on the centre of mass, and it is available for any data. Because U and the harmonic coordinates are both h-harmonic, it can be imposed at any radius. Section V shows that this mode dominated the error of every run until it was fixed.

III. THE PINN SOLVER

A. Networks

Inputs are the direction $n = x/\rho$ and the log-radial coordinate $t = \ln(\rho/\rho_{\rm in})/\ln(\rho_{\rm out}/\rho_{\rm in}) \in [0,1]$, optionally with the feature $\rho_{\rm in}/\rho$. The body is a tanh multilayer perceptron, typically of width 20 and depth 6. The output layer is initialised small ($\mathcal{N}(0, 10^{-2})$, zero bias), so that training starts near flat space. We compare two formulations.

First order: the network outputs h (6 components), Γ (18) and $\lambda = e^u$, and the compatibility equation $\partial_a h_{bc} = \Gamma^d{}_{ab} h_{dc} + \Gamma^d{}_{ac} h_{bd}$ is a residual like any other. Counting, compatibility (18) and Ricci (6) are 24 equations for 24 unknowns, with a three-parameter gauge degeneracy removed by (4).

Metric only: the network outputs h and λ, and Γ is the Christoffel symbol of h, obtained by automatic differentiation. Compatibility then holds identically, at the cost of second derivatives of the network output. Its residual stays at $\sim 3.5 \times 10^{-36}$ as a structural check.

For axisymmetric problems we use the ansatz

$$h_{ij} = (1+a)\,\delta_{ij} + b\,n_i n_j + \tfrac{c}{2}\,(n_i z_j + z_i n_j) + d\,z_i z_j, \qquad \lambda = e^u, \tag{11}$$

with a, b, c, d, u functions of $(t, \mu = n \cdot z)$. With width 20 and depth 6 this has 2265 parameters, or 2285 with the decay feature. The general three-dimensional metric-only network has 2347, and the first-order one 2725.

B. Loss

The residual groups are the Ricci equation (2), the gauge (4), the scalar equation (3) and, for first-order networks, compatibility. Each residual R_g is multiplied by ρ^{p_g}, and the loss is

$$L = \sum_g w_g\,\overline{(\rho^{p_g} R_g)^2} + w_{\rm in}\overline{r_{\rm in}^2} + w_{\rm out}\overline{r_{\rm out}^2} + w_{\rm pin}\sum\langle r\rangle^2 + w_{\rm com}\sum_S |d|^2, \tag{12}$$

with means over the sampled points of each group. The choice $p = (1,2,1,2)$ for (compat, Ricci, gauge, scalar) makes every residual dimensionless. With the features above, the loss is then invariant under a rescaling of the shell: the manufactured loss of a perturbed Weyl candidate is 1.720418×10^{-2} on the shells $[7.5, 75]$, $[0.1, 1]$ and $[0.01, 0.1]$ alike. For the quadrupole runs we use $p = (3,4,3,4)$, i.e. an extra weight ρ^4. With $\rho_{\rm in} = 1$ and $\rho_{\rm out} = 100$ the far rows then outweigh the inner ones by 10^8, and this is what determines the far field (Sec. V). Weights are $w_{\rm in} = w_{\rm out} = w_{\rm pin} = w_{\rm com} = 100$.

The centre-of-mass term uses a fixed quadrature on each sphere (Gauss–Legendre, 24 nodes in μ, times 8 uniform nodes in ϕ); a Monte Carlo estimate would be as noisy as the signal. Interior points are drawn log-uniformly in ρ with uniform directions: 16384 points in production, and 1024 on each boundary sphere. They are redrawn every 500 iterations, and every final result is checked on five fresh samples. The ratio of the fresh to the training loss lies between 0.94 and 1.06 for all results reported here.

C. Optimisers

All computations are in double precision with JAX [24]. A run consists of three stages.

1. Adam [25], 5000 steps.
2. A doubly-sketched Gauss–Newton method with trust region (DSGNAR) [26]. It sketches the residual Jacobian on rows (CountSketch) and columns (an SRCT embedding) to $s \times s$, with $s \approx 0.9$ of the parameter count, and solves Levenberg–Marquardt problems at 25 trust radii from one SVD. We run it in restart cycles, each with a fresh sample and a fresh trust region.
3. The self-scaled Broyden quasi-Newton method SSBroyden [27], with a dense inverse Hessian (0.04 GB at 2285 parameters), in the JAX implementation Crunch [28].

Table I and Fig. 1 compare the last two stages. A practical lesson concerns the first quasi-Newton step. With the inverse Hessian equal to the identity, the step $-\nabla L$ can be enormously too long. At a DSGNAR result with loss 8.5×10^{-5} the loss is NaN at the full step and 8.8×10^{6} at 1/16 of it, and Armijo backtracking needs 26 halvings to accept a step. An early version of the line search allowed 20, and its failure was the origin of several stalls that we first attributed to the problem. With a line search of up to 100 trials, a scale-invariant curvature guard, and the inverse Hessian carried between blocks of iterations, SSBroyden takes such a start from 8.1×10^{-5} to 1.1×10^{-10} in 16 min on one GPU.

Computations ran on NVIDIA A30 GPUs (24 GB) at the CCAD of the Universidad Nacional de Córdoba (cluster Mendieta, and a JupyterHub instance). One value-and-gradient evaluation with 16384 points on the 2285-parameter network takes 0.048 s on an A30, 24 times faster than on 96 CPU cores. For point counts that do not fit one GPU the interior points are sharded over the two GPUs of a node; the loss and gradient agree with the unsharded ones to 10^{-12} and 10^{-10}.

D. A spectral reference

For data without a closed-form solution we built an independent reference. It is the spherical background (6) plus an axisymmetric, reflection-even perturbation,

$$\delta\lambda = \sum_l f_0^l P_l, \qquad \delta h_{ij} = \sum_l \Big[f_1^l P_l\, \delta_{ij} + f_2^l P_l\, n_i n_j + f_3^l P_l'\,(e_i n_j + n_i e_j) + f_4^l P_l'' \Big(e_i e_j - \tfrac{1-\mu^2}{2}(\delta_{ij} - n_i n_j) \Big) \Big], \tag{13}$$

with $e = \hat{z} - \mu n$ and each f_k^l a Chebyshev series in $\ln\rho$. It is solved by Gauss–Newton on *exactly* the residual vector of the network's loss (12). The fine reference ($l \le 10$, 36 radial modes, 1008 unknowns) reaches a loss of 2.8×10^{-11}, and its far field falls off as $\rho^{-3.00}$ from $\rho = 20$ to 100. It is therefore a solution of the same discrete problem, obtained by an unrelated method.

IV. TESTS WITH EXACT SOLUTIONS

A. The pipeline on exact solutions

The exact solution (6), packed exactly as a network packs its outputs and fed through the same loss, gives a total loss below 10^{-16}. All four residual groups vanish to 10^{-16} or better. The rotated Weyl solution below gives compatibility 1.9×10^{-16}, Ricci 1.6×10^{-15} and scalar equation 7.1×10^{-17}, and the Weyl solution on the axis gives a total loss of 1.3×10^{-28} with the source (5). That test has teeth: with $S = 0$ the loss does not vanish, because the Weyl chart is not harmonic (the harmonic residual is 3.9×10^{-3} at $\rho = 7.5$).

B. Schwarzschild in harmonic coordinates

With Robin data at $\rho_{\text{out}} = 100$ and no manufactured source, the metric-only spherical network ($R_0 = 1$, $\lambda_0 = 0.381966$ on the sphere of h-areal radius 2) recovers the exact family. A fit of the trained solution gives $R_0 = 0.999491$ and $k = 1.000022$, and the maximum errors against the exact solution are 8.5×10^{-5} in h and 1.9×10^{-4} in Γ and λ. Table II shows the comparison that decided the formulation. On the same problem, the first-order network finds the same geometry in a different harmonic chart: λ as a function of the areal radius is right to 3.4×10^{-3}, but h_{rr} is off by 33%. The compatibility residual has a nearly flat direction along this gauge mode. Deriving Γ from h removes it.

C. Weyl two-rod solutions

The Weyl class [29] provides exact static axisymmetric solutions with nontrivial multipole structure. We use two rods, with potential $U_{\text{rod}} = \frac{1}{2}\ln[(R_a + R_b - L)/(R_a + R_b + L)]$ and mass $L/2$ each, and the metric function k obtained by quadrature. The data λ and the induced metric on the inner sphere, and the Robin sources, are taken from the exact solution, and the gauge is (5).

For two equal rods (masses 1, gap 1) on the shell $[7.5, 75]$, the axisymmetric network trained with SSBroyden from a random initialisation reaches a loss of 2.06×10^{-13} in 3000 iterations. The maximum errors over the shell are 6.7×10^{-7} in h and 9.0×10^{-8} in λ (Fig. 2). The decay powers of the Legendre modes of λ are -0.926, -2.05 and -2.85 for $l = 0, 1, 2$. With unequal masses (1 and 0.1) the solution has a dipole about the

TABLE I. SSBroyden against DSGNAR on the quadrupole problem of Sec. V (axisymmetric metric-only network, 2285 parameters). "Stalls" marks the first-step failure of the early line search discussed in the text.

setting	start	SSBroyden	DSGNAR
ρ^2, λ pin, from Adam	6.3×10^{-2}	1.74×10^{-8} (94 min)	9.4×10^{-8} (68 min)
ρ^4, λ pin	1.2×10^{4}	stalls	1.4×10^{-6} (88 min)
ρ^4, $\mathcal{P}$ pin	9.4×10^{-9}	2.1×10^{-9} (36 min)	3.4×10^{-9} (43 min)
+ centre of mass	0.395	stalls	1.16×10^{-10} (94 min)
+ centre of mass, polish	1.2×10^{-10}	3.75×10^{-11} (8 min)	3.96×10^{-11} (39 min)

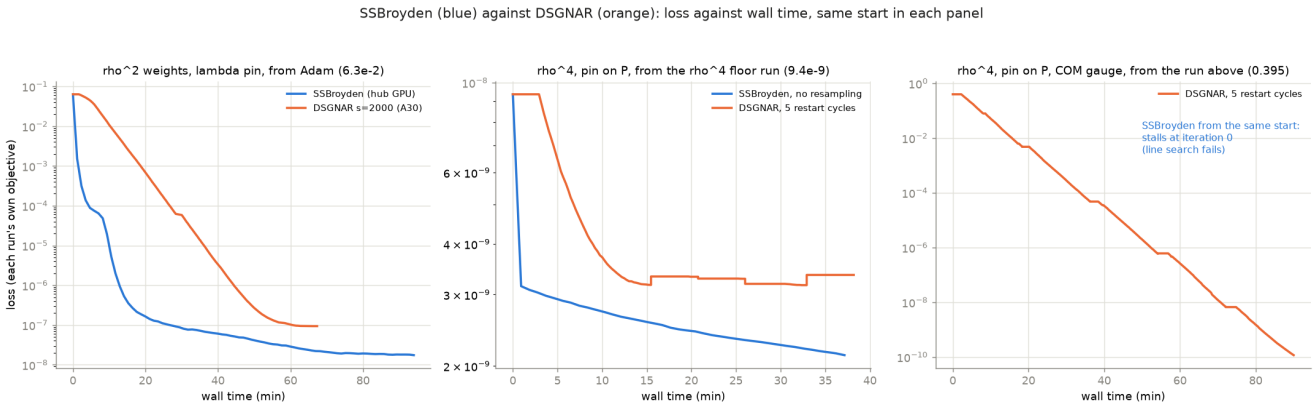


FIG. 1. Loss against wall time for SSBroyden (blue) and DSGNAR (orange) from the same start in each panel: ρ^2 weights from Adam; ρ^4 weights with the pin on $\mathcal{P}$; with the centre-of-mass condition. In the last panel SSBroyden made no step with the early line search; see the text. [TODO: regenerate without internal titles]

TABLE II. First-order against metric-only network on the same Schwarzschild problem (manufactured Robin source, $\lambda \to 1$, $\rho_{\text{out}} = 100$, 20000 Adam + 2000 quasi-Newton iterations).

	first order (h, Γ, λ)	metric only (h, λ)
compatibility rms	9.3×10^{-4}	1.7×10^{-9}
final loss	1.2×10^{-5}	5.8×10^{-8}
$\max \lvert\delta h\rvert$	3.3×10^{-1}	5.3×10^{-5}
$\max \lvert\delta \Gamma\rvert$	1.7×10^{-1}	2.3×10^{-4}
$\max \lvert\delta \lambda\rvert$	1.3×10^{-2}	2.9×10^{-5}

coordinate origin, and the network reproduces the multipole coefficients of λ at $\rho = 75$: $a_{00} = 3.44243418$ against the exact 3.44243419, $a_{10} = -1.017625 \times 10^{-3}$ against -1.017590×10^{-3}, and $a_{20} = -1.894 \times 10^{-5}$ against -1.898×10^{-5}. The dipole extracted at ρ_{out} is 1.398415 against the exact 1.398367. The Geroch–Hansen moments of the trained solutions, extracted as described in Sec. V D, are $M_0 = 2.000001$ and $M_2 = 2.49950$ for the equal rods (exact 2 and 2.5), and $M_0 = 1.100000$, $M_2 = 0.29105$, $M_3 = -0.5016$ with the centre of mass at $z = 1.3091$ for the unequal rods (exact 1.1, 0.290909, -0.49983 and 1.309091).

Bringing the inner sphere closer to the rods (Table III) raises the curvature at ρ_{in} by four orders of magnitude, and the error grows with it, but the solver keeps working at one horizon length from the rods.

TABLE III. Weyl rods (masses 1 and 0.1), inner sphere moved towards the rods. 3000 SSBroyden iterations; standard units.

shell	dist. to horizon	$R_{ab}R^{ab}(\rho_{\text{in}})$	$\max \lvert\delta h\rvert$	loss
$[7.5, 75]$	5.0	3.3×10^{-6}	3.5×10^{-7}	2.4×10^{-14}
$[4, 40]$	1.5	5.8×10^{-3}	1.6×10^{-5}	7.1×10^{-13}
$[3.5, 35]$	1.0	5.3×10^{-2}	7.7×10^{-5}	3.1×10^{-12}

D. A rotated Weyl configuration

Rotating the closest configuration by 45° in the z–x plane removes every symmetry. The equations are covariant, so $h \to RhR^T$ and $\Gamma \to R\,R\,R\,\Gamma$ (with λ unchanged), all evaluated at $R^T x$, is again a solution. The gauge, however, is a statement about a chart, and the source (5) must be taken about the rotated axis $n = R\hat{z}$. On the exact rotated solution the rotated source gives a gauge residual of 3.4×10^{-16}, while the harmonic residual is 3.0×10^{-2}. The general three-dimensional (first-order) network solves this problem, but the number of colloca-

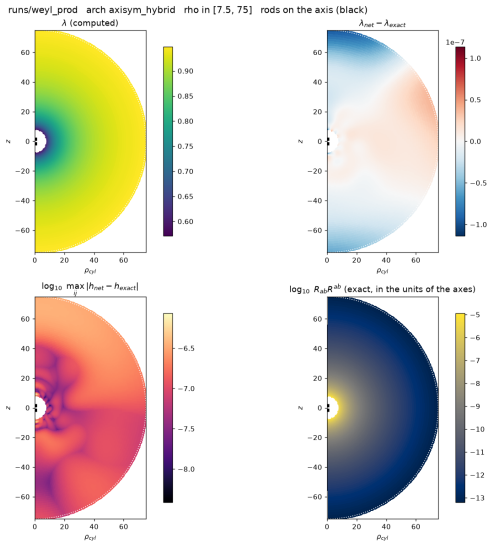


FIG. 2. Equal-mass Weyl rods on the shell $[7.5, 75]$ (meridional plane). Computed λ; $\lambda_{\rm net} - \lambda_{\rm exact}$; $\log_{10} \max_{ij} |h_{\rm net} - h_{\rm exact}|$; and $\log_{10} R_{ab}R^{ab}$ of the exact solution. [TODO: regenerate without the internal title]

TABLE IV. Rotated Weyl rods (masses 1 and 0.1, shell $[3.5, 35]$, 45°), general 3-D network.

	1500 it. 2048 pts	8000 it. 2048 pts	12165 it. 8192 pts	axisym.
loss	1.75×10^{-4}	2.28×10^{-5}	6.93×10^{-8}	3.06×10^{-12}
$\max \lvert\delta h\rvert$	7.1×10^{-2}	3.6×10^{-1}	1.1×10^{-3}	7.7×10^{-5}
$\max \lvert\delta\lambda\rvert$	5.2×10^{-3}	6.4×10^{-2}	1.0×10^{-4}	1.8×10^{-6}

tion points matters more than the number of iterations (Table IV). Running longer on 2048 points lowers the mean-square loss by a factor 13 while every error grows, because the optimiser concentrates the residual on a few points (the maximum of the scalar residual reaches 3375). Quadrupling the points lowers the loss by a factor 2530 and the maximum h error by 63. We draw two lessons: a mean-square loss needs enough points that the mean samples the spikes, and a plateau of the loss is no evidence of convergence until the errors are checked.

V. QUADRUPOLAR DATA

A. The problem

We take $R_0 = M = 1/\sqrt{3}$, the inner sphere $\rho_{\rm in} = 1$ with $r_{\rm in} = 1$ (so $\lambda_0 = 1/3$), $\rho_{\rm out} = 100$, and the datum

$$\lambda|_{\rho_{\rm in}} = \tfrac{1}{3} - \tfrac{1}{12}\, P_2(\cos\theta), \qquad (14)$$

a 25% relative perturbation of the spherical value. No closed-form solution is known. Linear theory around the spherical background predicts a pure $l = 2$ response that decays like ρ^{-3} in both λ and h, no $l = 1$ part, and an $l = 0$ back-reaction of order S_2^2/ρ^6. Any $l = 1$ content is therefore second order in S_2, numerical error, or a coordinate offset. We grade the networks against the fine spectral reference of Sec. III D. Against the spherical solution, the perturbation itself has an rms size of 2.0×10^{-2} in λ and 2.2×10^{-2} in h near the inner sphere, falling to 2.6×10^{-4} and 3.5×10^{-6} for $30 < \rho < 100$.

TABLE V. Without the centre-of-mass condition, the near-field error is the translation mode.

run	d_z	λ error, $1 < \rho < 3$	ratio
ρ^2, SSBroyden	-6.6×10^{-3}	4.1×10^{-4}	0.062
ρ^2, DSGNAR	$+1.4 \times 10^{-2}$	8.6×10^{-4}	0.061
ρ^4, $\mathcal{P}$ pin	-2.6×10^{-2}	1.6×10^{-3}	0.062
ρ^4, λ pin	-3.4×10^{-2}	2.1×10^{-3}	0.062
ρ^4, DSGNAR	-3.5×10^{-2}	2.1×10^{-3}	0.060

B. Weighting and the outer pin

With dimensionless weights ($p = (1, 2, 1, 2)$) the far field is under-determined: even the spectral reference has the correct ρ^{-3} tail of the P_2 mode only out to $\rho \approx 20$, and every network carries a flat $\sim 2 \times 10^{-4}$ error in λ beyond that. The ρ^4 weighting fixes the tail: the reference's P_2 slope becomes -3.00 from $\rho = 20$ to 70. It exposed, however, that the order-one pin on λ is violated by the exact solution itself. It leaves the $2M^2/\rho^2$ term, -6.6×10^{-5} at $\rho = 100$, which costs 4.3×10^{-7} in the loss: 98% of the loss of every ρ^4 run, a floor no optimiser could go below. Moving the pin to the potential (9) removes it (the same pin then costs 1.5×10^{-11}).

C. The translation mode and the centre-of-mass condition

With the weighting and the pin corrected, the networks' error near the inner sphere was still of order 10^{-3}, and it was *odd* under $z \to -z$ (modes $l = 1, 3$), which the even data forbid. Every term of the loss is exactly reflection symmetric, so nothing forced this part; it is a weakly controlled direction. The flux (10) identified it. In every run without the centre-of-mass condition, D_z is constant in ρ (e.g. -0.32 on the spheres $\rho = 1, 2, 10, 100$) but not zero, and the λ error in $1 < \rho < 3$ divided by $|d_z|$ is the same number in all of them, whatever the weighting, pin or optimiser (Table V). Far out, the network is the reference translated by $a = d/M = -0.0445$: translating the reference by a reduces the $l = 1$ difference by a factor 8–23 between $\rho = 10$ and 100.

Imposing $d = 0$ on the spheres $\rho = 1, 1.5, 3, 10, 30, 100$ removes the mode ($d_z \sim 10^{-8}$) and lowers the error

against the reference by four to five orders of magnitude everywhere in the shell (Table VI, Fig. 3). It is now five to six orders below the size of the perturbation. The loss falls with it, to 1.16×10^{-10} with DSGNAR and 3.75×10^{-11} after an SSBroyden polish, the level of the reference itself.

The condition carries over to data without reflection symmetry. With $S_1 = 0.05$ added to the datum (a genuine dipole in the data), the network reaches a loss of 8×10^{-10}. Against a spectral reference that includes all $l \le 10$, its rms error is 1.5×10^{-7} in λ and 1.9×10^{-6} in h, five orders below the perturbation.

D. Multipole moments

a. Extraction from the axis. For a static axisymmetric vacuum solution, the Geroch–Hansen moments [7, 8] follow from the potential on the symmetry axis in canonical Weyl coordinates, $g = -e^{2U}\mathrm{d}t^2 + e^{-2U}[e^{2k}(\mathrm{d}W^2 + \mathrm{d}Z^2) + W^2\mathrm{d}\phi^2]$. Writing $U(0,Z) = \sum_r U^{(r)}/|Z|^{r+1}$, the moments are [30] $M_0 = -U^{(0)}$, $M_1 = -U^{(1)}$, $M_2 = \frac{1}{3}(U^{(0)})^3 - U^{(2)}$ and $M_3 = (U^{(0)})^2U^{(1)} - U^{(3)}$ [31]. Equivalently, from the static Ernst potential $\xi = -\tanh U$ on the axis, $\xi(0,Z) = \sum_j m_j/|Z|^{j+1}$, the moments are $M_j = m_j$ for $j \le 3$ and $M_4 = m_4 - \frac{1}{7}m_0(m_0m_2 - m_1^2)$ [32]. We do not need to build Weyl coordinates, because only the axis is used. In our variables $U = \frac{1}{2}\ln\lambda$ and $W = |\partial_\phi|_h$. On the part of the axis that reaches infinity $k = 0$, so there $\mathrm{d}Z$ is the proper h-length along the axis. The inner ball cuts the axis in two; the offset between the branches is the change of Z along a meridian semicircle, with $\mathrm{d}Z = \star\mathrm{d}W$ in the meridian metric. We fit ξ (and U) on both branches and fix the origin of Z by mass centring, $M_1 = 0$. The error bar is the spread over fit windows starting at 1–1.5 ρ_{in} and fit orders 6 and 8. On exact solutions this returns:

- Schwarzschild in harmonic coordinates: $M_0 = R_0$ to 10^{-12}, and M_l below 10^{-11} for $1 \le l \le 4$;
- the equal Weyl rods: $M_0 = 2$ and $M_2 = 2.500000$;
- the unequal rods: the closed-form moments to 10^{-5} relative;
- Curzon: $M_2 = -M^3/3$ and $M_4 = 19M^5/105$, although U has no quadrupole.

b. The quadrupolar solution. Table VII lists the moments. The data raise the mass from the background value $R_0 = 0.57735$ to $M_0 = 0.583847$, a nonlinear back-reaction of 1.1% in the monopole, and the quadrupole moment is $M_2 = 0.1747$. The networks agree with the reference to 10^{-8} in M_0 and to 10^{-6}–10^{-8} in M_2. The run without the centre-of-mass condition, whose pointwise error was 10^{-3}, has moments within 10^{-5} of the reference. Moments are geometric, and the translation mode of Sec. II E moves the chart, not the geometry. The extraction shows this mode directly. The centre of mass of that run, measured from the centre of the coordinate sphere of radius ρ, is 10^{-5} at $\rho = 1$, then -0.020, -0.039 and -0.043 at $\rho = 2$, 10 and 70. It approaches the translation $a = -0.0445$ found from the flux and vanishes at the inner sphere, as the mode must. With the condition it is about 10^{-7} at every radius. The three-dimensional network gives moments within 10^{-3} relative of the reference, rotated or not.

c. Decay of the multipoles. The runs check during training that the tails decay at the order the equations require. Expanding $\mathcal{P}$ in Legendre modes $c_l(\rho)$ about the solution's own axis, the local power $\mathrm{d}\ln|c_l|/\mathrm{d}\ln\rho$ is -1.00, -3.00 and -5.0 for $l = 0, 2, 4$ in both the reference and the best network, the latter out to $\rho \approx 30$ where c_4 reaches 10^{-10}. The odd modes, which the data forbid, are at the 10^{-16} level in the reference and below 10^{-8} in the network. For the rotated three-dimensional network the $l = 2$ power is -3.00 out to $\rho = 30$ and drifts to -2.83 at $\rho = 100$, which is where its far-field error lies. In the mass-centred harmonic chart $c_l\rho^{l+1}$ tends to 0.58387, 0.1747 and 0.0220, i.e. to M_0, M_2 and M_4. This is consistent with harmonic coordinates being asymptotically Cartesian and mass-centred coordinates [9, 33], and it gives a second, independent route to the same numbers.

E. Off-axis data and the general ansatz

Rotating the quadrupolar datum to the axis $(1,0,1)/\sqrt{2}$ makes the problem fully three-dimensional, while its exact solution is the rotated axisymmetric one, which the rotated spectral reference provides. With the general metric-only network (2347 parameters), the error against the rotated reference is 1.6×10^{-5} in λ and 1.2×10^{-3} in h (rms). That is 600 and 10 times below the perturbation, and the same as for the same network on the unrotated data (1.65×10^{-5} and 1.0×10^{-3} after the same budget), so the rotation itself costs nothing. The general ansatz, however, floors at a loss between 10^{-7} and 10^{-6}, depending on the number of points: three to four orders above the axisymmetric network on the same objective (1.1×10^{-10}). Three diagnostics locate the limit. (i) The reference, evaluated with the network's own loss on fresh samples, gives 2.8×10^{-11}, so the minimum of the objective is at the solution. (ii) The same network, fitted by regression to the reference, represents it to 7.5×10^{-6} in h, 150 times better than training reaches, but with a PINN loss of 3.4×10^{-2}. (iii) More collocation points (up to 262144) lower the loss to 1.2×10^{-7} but leave the h error between 7×10^{-4} and 10^{-3}. Training started from the regression fit reaches a loss of 2×10^{-7} with an h error of 3.2×10^{-4}, the best we obtained. We conclude that there is a direction in h, concentrated near the inner sphere, along which the residual is small and the error is not, and that for the general ansatz the optimiser drifts along it. [TODO: identify the mode (angular decomposition of the error).]

TABLE VI. Error against the fine reference (angular rms) by radial region, one modification at a time. Rows 0–2 have no centre-of-mass condition.

	loss	λ error				h error			
		1–3	3–10	10–30	30–100	1–3	3–10	10–30	30–100
perturbation size		2.0×10^{-2}	2.5×10^{-3}	7.6×10^{-4}	2.6×10^{-4}	2.2×10^{-2}	4.5×10^{-4}	3.1×10^{-5}	3.5×10^{-6}
0: ρ^2, λ pin	1.7×10^{-8}	4.1×10^{-4}	2.5×10^{-4}	2.1×10^{-4}	2.2×10^{-4}	5.1×10^{-3}	7.5×10^{-4}	9.2×10^{-5}	2.9×10^{-5}
1: ρ^4, λ pin	4.5×10^{-7}	2.1×10^{-3}	1.0×10^{-3}	1.6×10^{-4}	1.9×10^{-5}	2.6×10^{-2}	3.9×10^{-3}	3.7×10^{-4}	3.9×10^{-5}
2: ρ^4, $\mathcal{P}$ pin	2.1×10^{-9}	1.6×10^{-3}	7.8×10^{-4}	1.2×10^{-4}	1.4×10^{-5}	2.0×10^{-2}	2.9×10^{-3}	2.8×10^{-4}	2.9×10^{-5}
3: + centre of mass	1.2×10^{-10}	6.7×10^{-8}	3.4×10^{-8}	1.2×10^{-8}	3.4×10^{-9}	4.8×10^{-7}	3.7×10^{-8}	5.0×10^{-9}	4.0×10^{-9}
3 + SSBroyden polish	3.8×10^{-11}	3.2×10^{-8}	3.9×10^{-9}	2.9×10^{-9}	2.6×10^{-9}	1.4×10^{-7}	1.7×10^{-8}	2.4×10^{-9}	1.6×10^{-9}

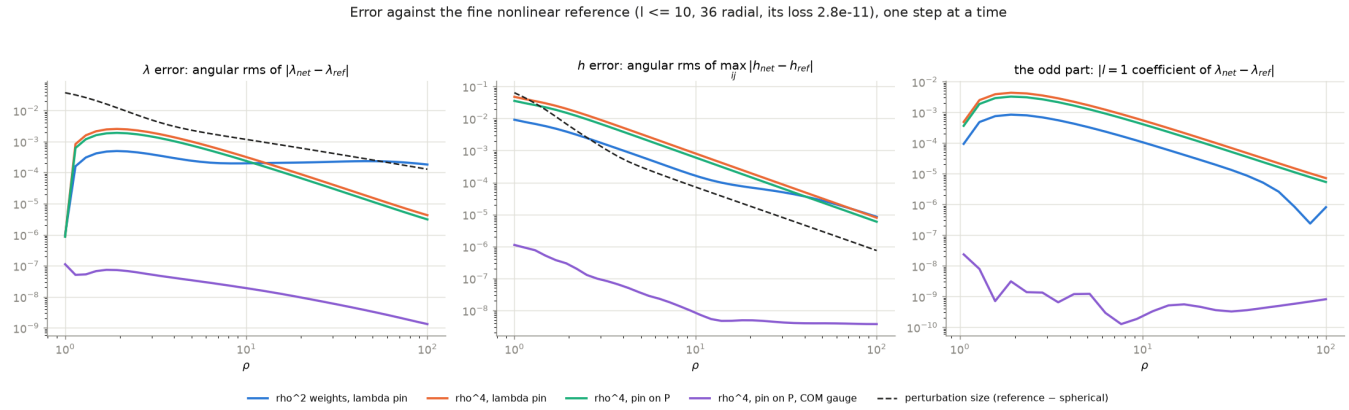


FIG. 3. Error against the fine reference for the steps 0–3 of Table VI: angular rms of $\lambda_{\mathrm{net}} - \lambda_{\mathrm{ref}}$, of $\max_{ij} |h_{\mathrm{net}} - h_{\mathrm{ref}}|$, and the $l = 1$ coefficient of $\lambda_{\mathrm{net}} - \lambda_{\mathrm{ref}}$. Dashed: size of the perturbation (reference minus spherical solution). [TODO: regenerate without the internal title]

TABLE VII. Geroch–Hansen moments of the quadrupolar solution: the spectral reference, and the differences network − reference (same fit windows). The window spread of the reference is 1.2×10^{-4} in M_2 and 7×10^{-4} in M_4; it largely cancels in the differences.

	M_0	M_2	M_4
reference	0.583847	0.17471	0.0220
axisym., centre of mass	-2.8×10^{-8}	$+1.1 \times 10^{-6}$	-1.0×10^{-5}
axisym., workflow from scratch	-1.0×10^{-8}	$+2.5 \times 10^{-8}$	$+1.1 \times 10^{-6}$
axisym., no centre of mass	$+1.0 \times 10^{-7}$	$+1.2 \times 10^{-5}$	-1.3×10^{-4}
3-D, data on the axis	-2.3×10^{-6}	-3.3×10^{-5}	-1.9×10^{-4}
3-D, data rotated	$+2.2 \times 10^{-6}$	$+1.3 \times 10^{-4}$	-1.9×10^{-4}

VI. CONCLUSIONS

Physics-informed neural networks solve the exterior boundary-value problem for the static vacuum Einstein equations with prescribed inner data and asymptotic flatness. They are accurate to 10^{-7}–10^{-9} against exact and reference solutions when four ingredients are in place:

- a metric-only network, in which the connection is derived from the metric;
- dimensionless or far-weighted residuals with Robin conditions of the correct orders;
- a quasi-Newton or sketched Gauss–Newton optimiser;
- a complete gauge fixing.

The last point was not anticipated. Harmonic coordinates together with the inner data leave the asymptotic chart free to translate. The mode is invisible to every residual except at the 10^{-9} level, and it was the largest error in every run until it was fixed by a centre-of-mass condition based on a conserved flux of the static equations. We expect the same issue, and the same remedy, in any elliptic formulation of the stationary problem in which the inner data do not fix the asymptotic frame.

The Geroch–Hansen moments of the computed solutions can be read off the symmetry axis without constructing Weyl coordinates; for the quadrupolar data they agree with the reference to 10^{-8}–10^{-4}, and they are insensitive to the chart ambiguity that dominated the pointwise error. An open question is the accuracy floor of the general three-dimensional ansatz, which we traced to a weakly penalised direction of the objective rather than to representation or sampling. The extension to the stationary equations is the natural next step.

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